

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	SWTID-2026-3278
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The project aims to develop a **Credit Card Approval Prediction System** using Machine Learning. The system analyzes applicant details such as income, employment status, credit score, age, loan history, and existing debt to predict whether a credit card application should be approved or rejected. This helps financial institutions make faster, more accurate, and unbiased decisions while reducing manual effort and fraud.

Project Overview	
Objective	To automate the credit card approval process using machine learning, enabling fast, accurate, and reliable approval decisions.
Scope	The project predicts credit card approval based on customer financial and personal information. It assists banks and financial institutions in reducing processing time, minimizing risk, and improving customer satisfaction.
Problem Statement	
Description	Traditional credit card approval is often slow and depends on manual verification, leading to delays, human errors, and inconsistent decisions.
Impact	An intelligent prediction system improves approval accuracy, reduces processing time, minimizes financial risk, and enhances the overall customer experience.

Proposed Solution	
Approach	Develop a machine learning model that predicts credit card approval using applicant information such as income, credit score, employment status, debt-to-income ratio, age, and loan history.
Key Features	• Automated approval prediction • High prediction accuracy • Fast decision-making • Fraud risk detection • Easy-to-use interface • Data-driven recommendations



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	GPU (Optional)	NVIDIA T4 or equivalent
Memory	RAM	8 GB
Storage	Space for dataset, models, and project files	256 GB SSD (500 GB–1 TB preferred)
Software		
Frameworks	Machine Learning Frameworks	Scikit-learn, Pandas, NumPy, Flask
Libraries	Additional libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Joblib

IDE	Development Environment	Jupyter Notebook or VS Code
Data		
Data	Source, size, format	Kaggle/UCI Repository, CSV format, approximately 600–1000 records